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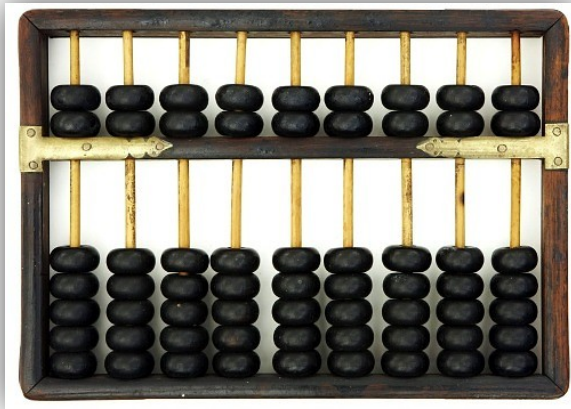
HISTORY OF COMPUTERS

Hello, my name is Adrian Tennant and I will be talking about the importance of being knowledgeable about technology and life; utilizing the history of Computers. You are expected to learn at least: 1) What a computer is? and 2) How they changed over time and how they have influenced your life today.

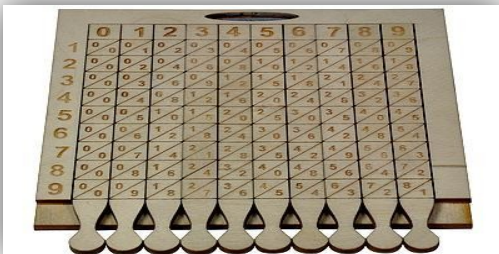
The Origin of Computers started a long time ago and the computers in the past differ quite a lot from the computing devices we are used to today; Let's start off with what is a computer – it can be defined as any device which assist humans in executing diverse computations or calculations. Persons used sticks, stones, bones and humans as counting tools before modern computers were invented, in fact individuals whose jobs were to carry out calculations were called computers the term “computer” was first documented in 1613 referring to such individuals. As time progressed and increase in the quantity of computing devices were produced as technology developed and the human ingenuity got enhanced.

Here is an outlined detail of the history of computers starting from Early computing devices and ending at the ones we have today:

THE ABACUS



An abacus is a mechanical device used to aid an individual in performing mathematical calculations it originated in Babylonia in 2400 B.C. Its name was derived from the Latin word **abakos**, which means “calculating table.” It consists of a frame with rods and movable beads and is considered as the first computer. The abacus in the form we are most acquainted with was first used in China in around 500 B.C. It used to perform basic arithmetic operations. [1, 2, 3]



NAPIERS BONES

This was invented by John Napier in 1614.

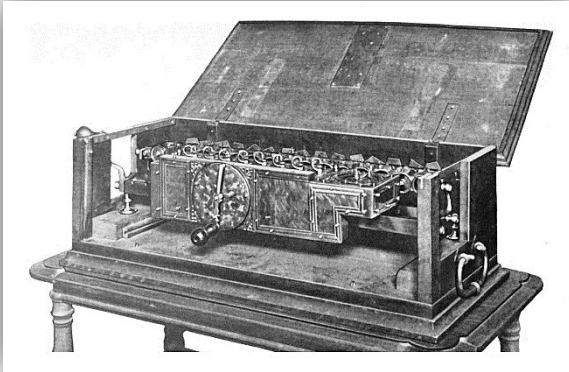
It is more complicated abacus and enable and individual to user to multiply, divide and calculate square and cube roots by manipulating rods and putting them on uniquely made boards. [1, 2]

PASCAL CALCULATOR



This was invented by Blaise Pascal in 1642 he was engaged in mathematics and philosophy and made this an attempt to make taxes easier for his father. It was quite costly. [1, 2, 4]

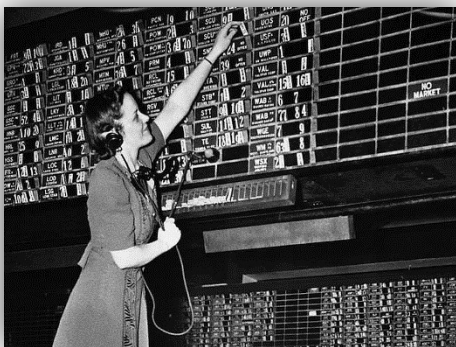
STEPPED RECKONER



This was made by Gottfried Wilhelm Leibniz in the year 1672. It can perform the four operations of mathematics automatically. [1, 2, 5]

After these then came the generations, there were notably 5 generations of computers which I will share below:

THE FIRST GENERATION - Vacuum Tubes (1940–1956)



These first and expensive in terms of cost and usage computers used vacuum tubes for circuitry and magnetic drums for memory, and were often enormous, taking up entire rooms and often need A/C. They relied heavily on the use of machine language and was capable of processing one problem at a time. Input was based on punched cards and paper tape, and output was displayed on printouts. An example of this computer is IBM-650 (650 Magnetic Drum Data-Processing Machine). [1, 6, 7]



THE SECOND GENERATION- Transistors (1959–1965)



Transistors replaced vacuum tubes and which motivated the emergence of the second generation of computers. One transistor was equal 40 vacuum tubes which enabled computers to become littler and

more cost effective and overall more dependable. But there exists the problem of the computer over heating due to this new technology. Here also introduced the shift from cryptic binary machine language to assembly and used to magnetic core technology to store instructions. An example of this computer is the UNIVAC 1108. [1, 6, 8]

THE THIRD GENERATION- Integrated Circuit (1964–1974)



These computers were made using integrated circuits. Transistors got smaller and they were placed on their circuit boards along with silicone chips/semiconductors made gave a massive performance boost in speed and proficiency over the previous generations these computers also used interfaced operating systems. Computers could now carry out instructions at never before seen speeds. ex a billionth of second. They became more accessible to the public because of the drastic decrease in price and size. And the use of peripherals such as



keyboard and monitors started with these computers as

well. An example of this computer is the IBM-360 series. [1, 6, 9]



THE FOURTH GENERATION- **MICROPROCESSORS (1971–1980)**

The emergence of the microprocessors facilitated this generation of computers. As a result of innovations in computers architecture to implements of computer networks started then the formation of the internet followed suit. Graphics user interfaces were also made and creation of mice and other handheld happened. An example of a 4th gen computer is the: CRAY-1(Super Computer). [1, 6, 10, 11]

THE FIFTH GENERATION- **ARTIFICIAL INTELLIGENCE (1980-** **Present)**



These are systems embedded with artificial intelligence and or robotics, these are improving day by day at a rapid pace. This is the generation we are in presently. ULSI (Ultra Large Scale Integration) technology is prominent in this generation. This generations computers have ten million electronic components, resulting in the production of microprocessors chips. The purpose of this generation is make computers more humanized and so that they can respond to natural language. example this I today facial and voice recognition in mobile devices. Higher level programming languages are used in like C++, .Net and Java are widely used in this generation. And almost everyone has a 5th gen computer including you.

Examples of these computers include:

- **MICROCOMPUTERS-** laptops, tablets, PDAs, PCs.
- **MINICOMPUTERS-** IBM36, HP9000.
- **MAINFRAMES** - System z9 server and System z10 servers.
- **SUPERCOMPUTERS** - Tianhe-I, IBM Roadrunner. [12, 1, 6]

THE CONCLUSION

Here concludes this document; we have explored how innovations in computing devices happened over the years and the technology used in facilitating its development. We sure have come a far way from using analog and mechanical systems to more sophisticated electrical circuitry and AI technology to solve problems, I hope you discovered something new from my research. [6]

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